

BASICS FOR AIR TRAFFIC CONTROL – SIDs AND STARs

INTRODUCTION

Air traffic in the vicinity of airports naturally becomes congested. Much like for automobile traffic, specialized procedures have been developed to organize and expedite the flow of aircraft into and out of terminal areas. These complicated and intricate procedures have been encoded on:

- Standard Instrument Departure (SID) charts
- Standard Terminal Arrival (STAR) charts

The purpose of this module is to introduce you to SID and STAR procedural charts, which are used by pilots when flying on an Instrument Flight Rules (IFR) flight plan.



General Disclaimer: All graphics in this lesson are for illustration/training purposes only and may not reflect current procedures.

STANDARD INSTRUMENT DEPARTURE (SID) CHARTS

Purpose: The purpose of this lesson is to describe how Standard Instrument Departure (SID) procedures are used and to identify the types of information displayed on these charts.

Objectives:

- Identify types of departure procedures
- Identify types of SIDs
- Identify components of SIDs
- Identify “No SID” procedures

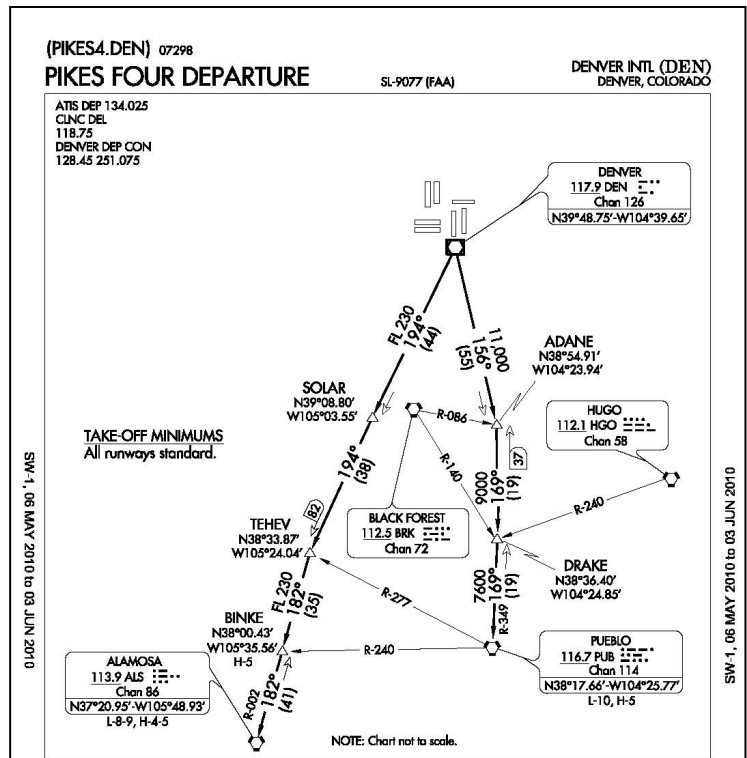
References for this lesson are as follows:

- FAA Order JO 7110.65, Air Traffic Control
- FAA-H-8083-16, Instrument Procedures Handbook
- U.S. Terminal Procedures Chart SW, Vol. 1
- Aeronautical Information Manual (AIM)

Standard Instrument Departures (SIDs)

A SID is a departure procedure used by the pilot/controller to **provide obstacle clearance and a transition from the terminal area to the appropriate en route structure.**

They are provided to pilots in graphical and textual form within **US Terminal Procedures Publications.**

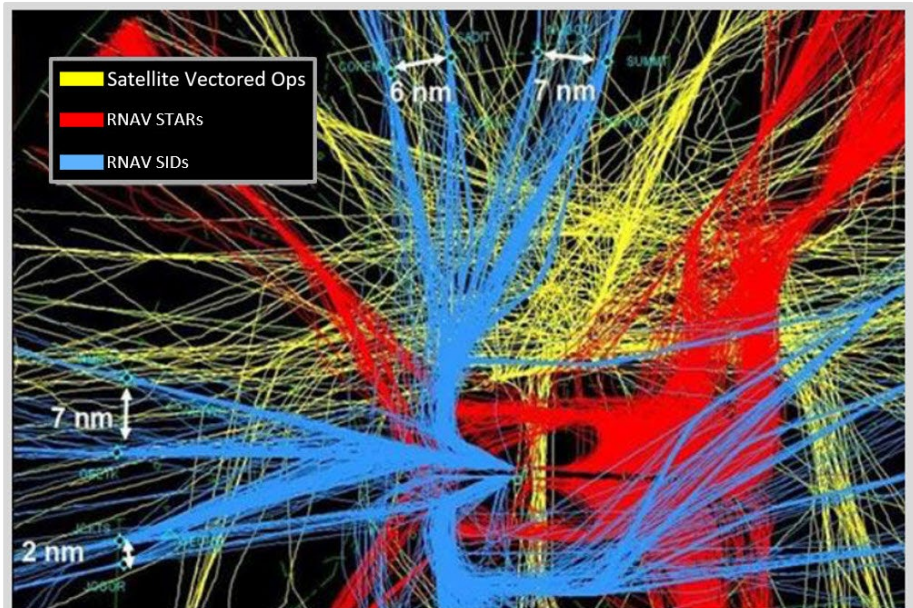


Performance-Based Navigation (PBN)

Performance-Based Navigation (PBN) is a **satellite navigation tool that creates precise 3D flight paths.** PBN helps **lower aircraft fuel consumption and emissions** and **saves time.** It incorporates satellites with our ground-based navigational systems.

PBN impacts departures:

- Reduces controller workload
- Provides for more departures per hour while using reduced departure procedures
- Reduces dependence on existing navigational aids

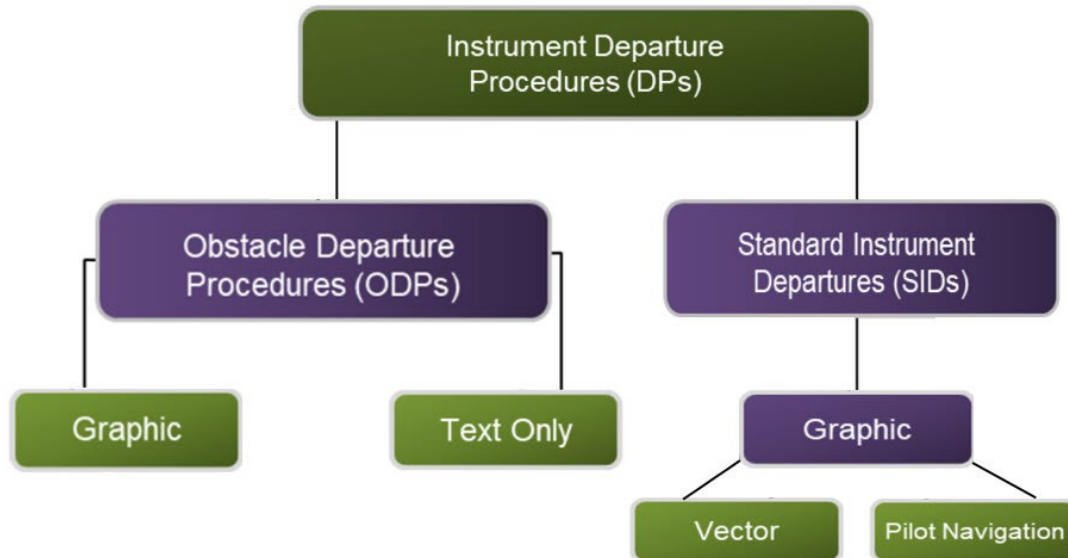


This diagram depicts RNAV arrival and departure flight paths at a busy airport.

Instrument Departure Procedures (DPs)

Instrument Departure Procedures (DPs) are a preplanned instrument flight rule (IFR) departure procedure published for pilot use, in graphic or textual format, that provides obstruction clearance from the terminal area to the appropriate en route structure.

There are two types of DPs: **Obstacle Departure Procedure (ODP)** and Standard Instrument Departure (SID).



Obstacle Departure Procedures (ODPs)	Standard Instrument Departure (SIDs) Procedures
Obstacle Departure Procedures (ODPs) are preplanned IFR departure procedures printed for pilot use in textual or graphic form to provide obstruction clearance via the least onerous route from the terminal area to the appropriate en route structure. ■ ODPs are recommended for obstruction clearance and may be flown without ATC clearance unless an alternate departure procedure (SID or radar vector) has been specifically assigned by ATC ■ ODPs are recommended when no SID or radar vector is assigned because they provide obstruction clearance via the least difficult route	Standard Instrument Departures (SIDs) Procedures are preplanned IFR ATC departure procedures printed for pilot/controller use in graphic form to provide obstruction clearance and a transition from the terminal area to the appropriate en route structure. ■ SIDs are primarily designed for system enhancement to expedite traffic flow and to reduce pilot/controller workload ■ ATC clearance must always be received prior to flying a SID Note: This lesson will cover SIDs, which are the most common type of DPs used by ATC. Not all airports have an Instrument DP.

Standard Instrument Departure (SID) Charts

To find a particular SID in the terminal procedures publications, locate the city first, then the airport serving that city.

- SIDs are always located after the Instrument Approach Procedures (IAPs)
- One airport may have several SIDs

INDEX			INDEX OF TERMINAL CHARTS AND MINIMUMS		
NAME	PROC	SECT PG	NAME	PROC	SECT PG
DENVER, CO (CONT)			DENVER, CO (CONT)		
DENVER INTL (DEN)			FRONT RANGE (FTG)		
TAKE-OFF MINIMUMS	C	22	TAKE-OFF MINIMUMS	C	22
ALTERNATE MINIMUMS	E	22	STARS - DANDO FIVE		22
STARS - DANDO FIVE		22	LANDR FIVE		28
LANDR FIVE		28	LARKS FIVE		28
LARKS FIVE		28	POWDR SEVEN		212
POWDR SEVEN		212	QUAIL FIVE		214
QUAIL FIVE		214	RAMMS FIVE		215
RAMMS FIVE		215	SAYGE SIX		217
SAYGE SIX		217	TOMSN FOUR		219
TOMSN FOUR		219	IAPS - ILS OR LOC RWY 25		161
IAPS - ILS OR LOC RWY 8		118	ILS RWY 17		162
ILS OR LOC RWY 16R		119	ILS RWY 36		163
ILS OR LOC RWY 17L		120	RNAV (GPS) RWY 17		164
ILS OR LOC RWY 34L		121	RNAV (GPS) RWY 26		165
ILS OR LOC RWY 7		122	RNAV (GPS) RWY 35		166
ILS OR LOC RWY 25		123	NDB RWY 26		167
ILS RWY 16L		124	AIRPORT DIAGRAM		168
ILS RWY 17R		125	DPS - DENVER FIVE		169
ILS RWY 26		126	PIKES FOUR		171
ILS RWY 34R		127	PLAINS FOUR		172
ILS RWY 35L		128	ROCKIES SIX		174
ILS RWY 36R		129	YELLOWSTONE FIVE		176
ILS RWY 36L		130	ROCKY MOUNTAIN METROPOLITAN (BJC)		
ILS RWY 36R (B)		131	TAKE-OFF MINIMUMS	C	
ILS RWY 36L (B)		132	ALTERNATE MINIMUMS	E	
ILS RWY 36R (B)		133	STARS - DANDO FIVE		22
ILS RWY 36L (B)		134	LANDR FIVE		28
ILS RWY 36R (B)		135	LARKS FIVE		28
ILS RWY 36L (B)		136	POWDR SEVEN		212
ILS RWY 36R (B)		137	QUAIL FIVE		214
ILS RWY 36L (B)		138	RAMMS FIVE		215
ILS RWY 36R (B)		139	SAYGE SIX		217
ILS RWY 36L (B)		140	TOMSN FOUR		219
ILS RWY 36R (B)		141	IAPS - ILS OR LOC RWY 25R		178
ILS RWY 36L (B)		142	GPS RWY 29L		179
ILS RWY 36R (B)		143	GPS RWY 29R		180
ILS RWY 36L (B)		144	VOR/DME RWY 29L/R		181
ILS RWY 36R (B)		145	AIRPORT DIAGRAM		182
ILS RWY 36L (B)		146	DPS - DENVER FIVE		183
ILS RWY 36R (B)		147	PIKES FOUR		185
ILS RWY 36L (B)		148	PLAINS FOUR		186
ILS RWY 36R (B)		149	ROCKIES SIX		188
ILS RWY 36L (B)		150	YELLOWSTONE FIVE		190
ILS RWY 36R (B)		151			
ILS RWY 36L (B)		152			
ILS RWY 36R (B)		153			
ILS RWY 36L (B)		154			
ILS RWY 36R (B)		155			
ILS RWY 36L (B)		156			
ILS RWY 36R (B)		157			
ILS RWY 36L (B)		158			
ILS RWY 36R (B)		159			
ILS RWY 36L (B)		160			
ILS RWY 36R (B)		161			
ILS RWY 36L (B)		162			
ILS RWY 36R (B)		163			
ILS RWY 36L (B)		164			
ILS RWY 36R (B)		165			
ILS RWY 36L (B)		166			
ILS RWY 36R (B)		167			
ILS RWY 36L (B)		168			
ILS RWY 36R (B)		169			
ILS RWY 36L (B)		170			
ILS RWY 36R (B)		171			
ILS RWY 36L (B)		172			
ILS RWY 36R (B)		173			
ILS RWY 36L (B)		174			
ILS RWY 36R (B)		175			
ILS RWY 36L (B)		176			
ILS RWY 36R (B)		177			
ILS RWY 36L (B)		178			
ILS RWY 36R (B)		179			
ILS RWY 36L (B)		180			
ILS RWY 36R (B)		181			
ILS RWY 36L (B)		182			
ILS RWY 36R (B)		183			
ILS RWY 36L (B)		184			
ILS RWY 36R (B)		185			
ILS RWY 36L (B)		186			
ILS RWY 36R (B)		187			
ILS RWY 36L (B)		188			
ILS RWY 36R (B)		189			
ILS RWY 36L (B)		190			
ILS RWY 36R (B)		191			
ILS RWY 36L (B)		192			
ILS RWY 36R (B)		193			
ILS RWY 36L (B)		194			
ILS RWY 36R (B)		195			
ILS RWY 36L (B)		196			
ILS RWY 36R (B)		197			
ILS RWY 36L (B)		198			
ILS RWY 36R (B)		199			
ILS RWY 36L (B)		200			

Types of SIDs

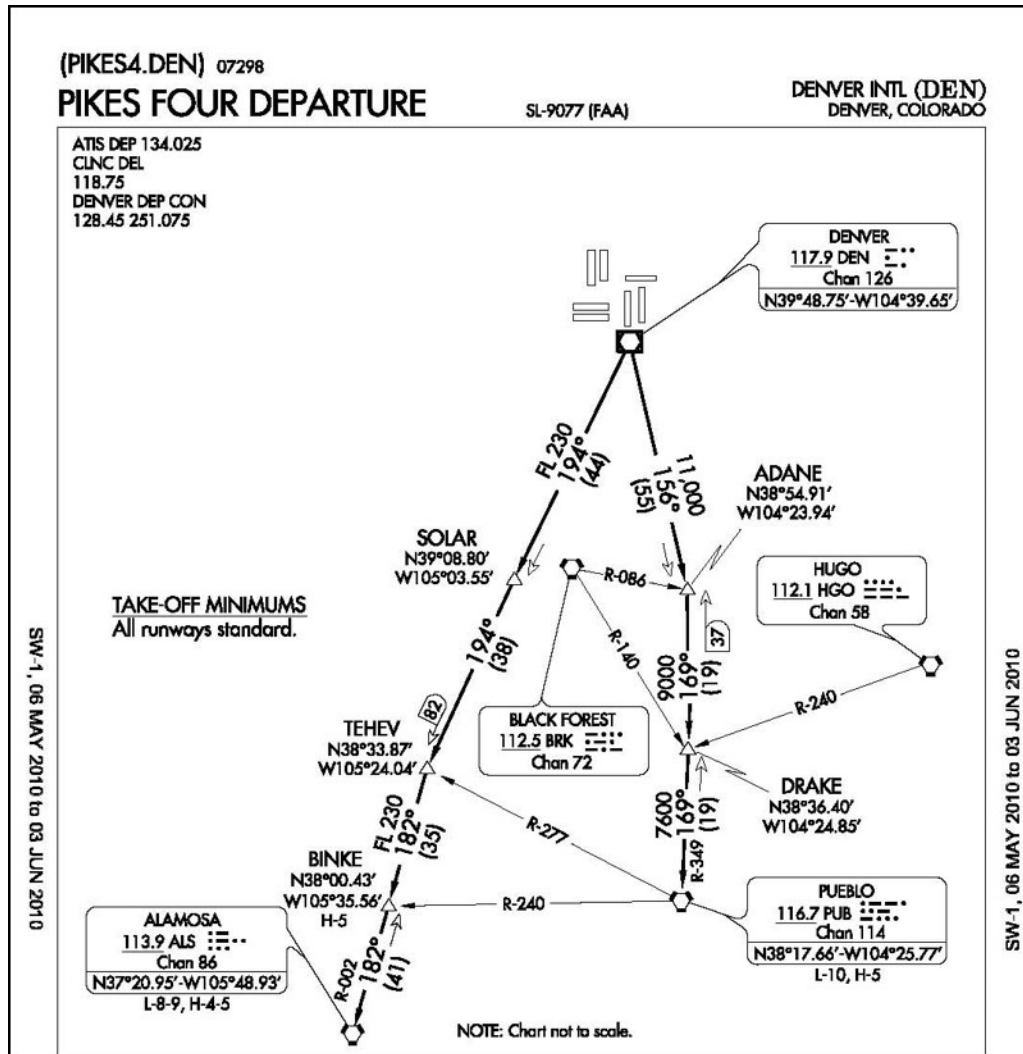
There are two basic types of SIDs:

Pilot Navigation SID	Vector SID
Pilot navigation SIDs are designed to allow pilots to provide their own navigation with minimal radio communication. ■ May include an initial segment requiring radar vectors to help the flight join the procedure	Vector SIDs require ATC to provide radar vectors from just after takeoff until reaching the assigned route or a fix depicted on the SID chart. ■ Does not include departure routes or transition routes because independent pilot navigation is not involved

Pilot Navigational SID

Pilot Navigation SIDs use **precise radials from a Very High Frequency Omnidirectional Range (VOR)** that have been flight checked by FAA pilots to ensure accuracy.

A pilot navigation SID may include an initial segment requiring radar vectors to help the flight join the procedure.



Vector SID

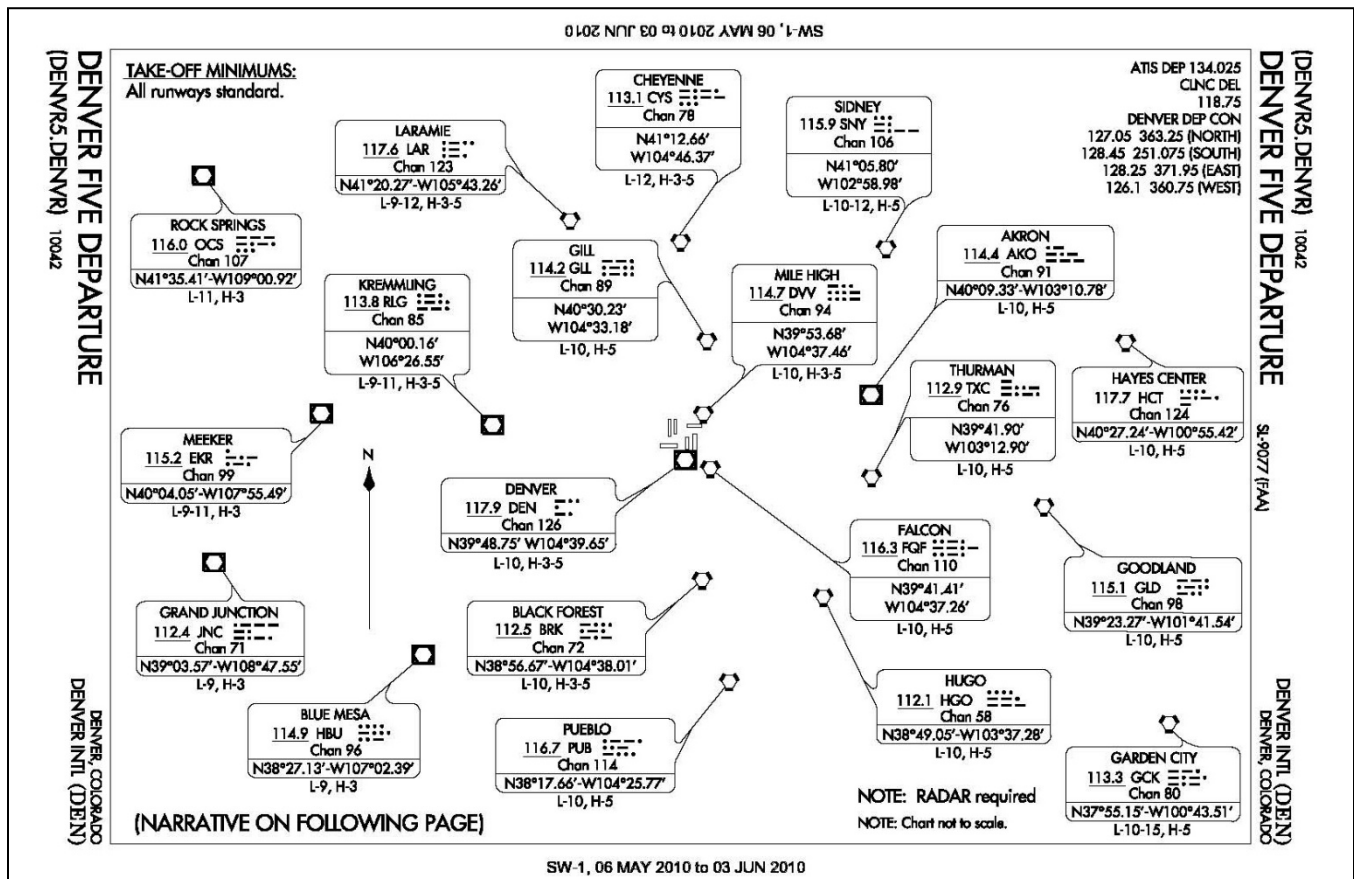
Vector SIDs do not include departure routes or transition routes because independent pilot navigation is not involved.

The procedure sets forth an initial set of departure instructions that typically include an initial heading and altitude.

ATC must have radar contact with the aircraft to be able to provide vectors.

The pilot is expected to immediately comply with a radar vector and to notify ATC if they are unable to comply.

The pilot is also expected to make contact immediately if an instruction will cause them to compromise safety due to obstructions or traffic.



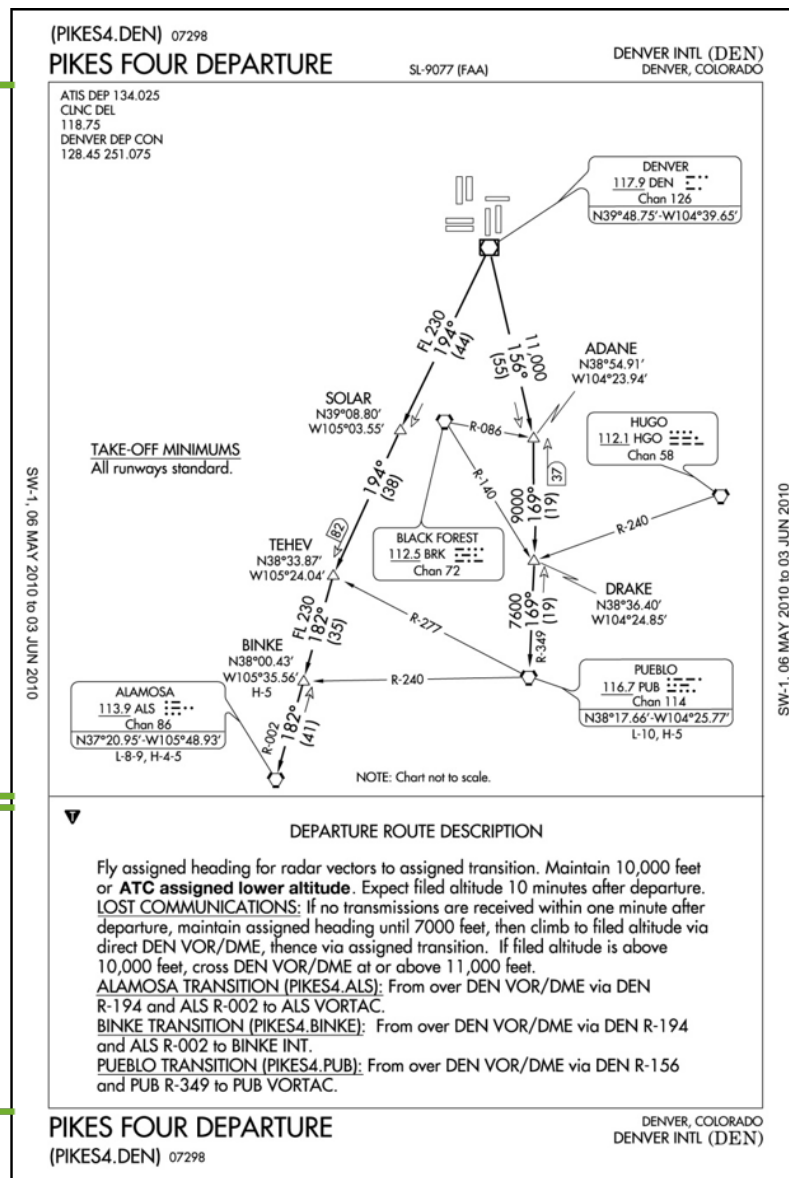
SID Layout

The three sections of a SID are:

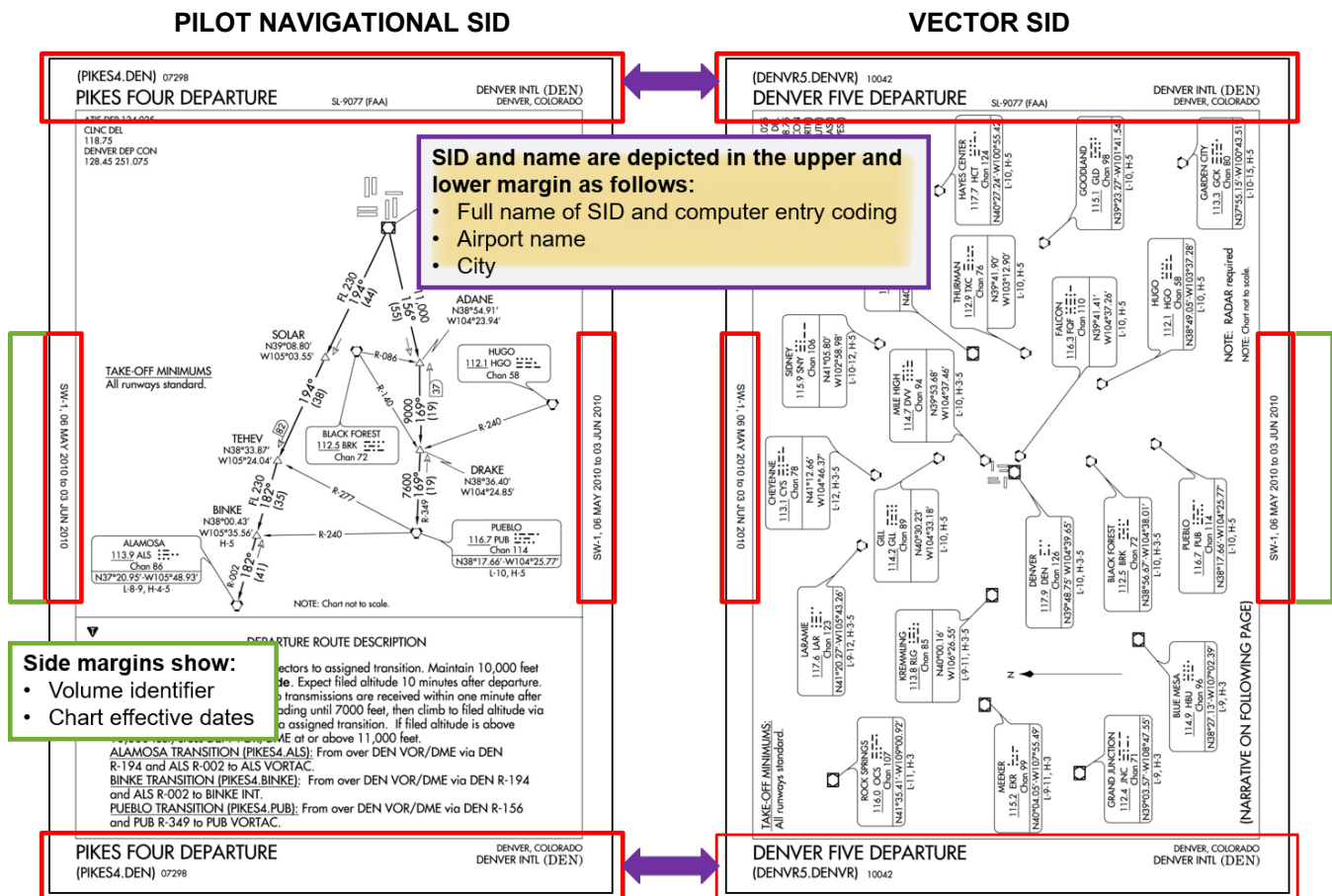
Planview

Textual
Description

Margin
Information



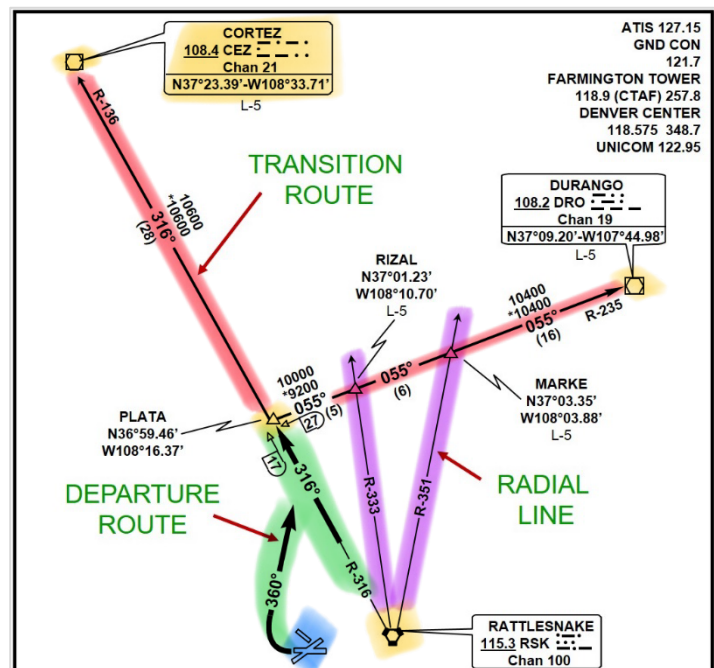
Margin Information



Planview

The Planview (graphic) contains many symbols which have already been studied, including:

- Airport runways
- NAVAIDs such as VORs, VORTACs, and intersections
- Primary NAVAID boxes
- Communication frequency section
- Departure routes
- Transition route
- Radials of departure routes
- Distance between fixes
- Minimum en route altitude (MEA)



The Textual Description is a narrative that correlates with the Planview. The pilot is required to have the appropriate chart of the SID at the time of departure.

“No SID”

- Cannot comply with a SID
- Do not possess SID charts
- Simply do not wish to fly a SID

Although it is not required, pilots are **encouraged** to include the phrase, “No SID,” in the remarks portion of their flight plan. The flight progress strip will reflect this decision.

Flight Progress Strip

[illegible]

When a pilot indicates they will not accept a SID, the controller should clear the aircraft via the filed route to the extent possible, or via a preferential departure route. *Pilot cannot accept a SID*

A **preferential departure route (PDR)** is a specific departure route from an airport or terminal area to an en route point where there is no further need for flow control.

Example

The underlined segments depict the difference between the example clearances.

A clearance without using a SID may sound like:	The same clearance using a SID would sound like: ❤
"United Six Fifty Five, cleared to Burbank airport, <u>after departure turn right heading 360, intercept Rattlesnake 316 radial direct Plata intersection, then via Durango 235 radial direct Durango</u> , then as filed. Climb and maintain flight level 210 departure frequency 119.1 squawk 0454."	"United Six Fifty Five, cleared to Burbank airport, <u>Plata 1 departure, Durango transition</u> , then as filed. Climb and maintain flight level 210 departure frequency 119.1 squawk 0454."



Knowledge Check A

REVIEW what you have learned so far about SID charts. ANSWER the questions listed below.

- Which procedures are preplanned IFR procedures that provide obstruction clearance and facilitate transition from the terminal environment to the en route environment? (Select the correct answer.)
 - ☐ Standard Instrument Departures
 - ☐ Obstacle Departure Procedures
 - ☒ Instrument Departure Procedures
- Which of the following should be located first in the Terminal Procedures Publications when searching for a particular SID? (Select the correct answer.)
 - ☐ Airport
 - ☒ City
 - ☐ State
- What should the controller do when a pilot indicates that they will not accept a SID? (Select the correct answer.)
 - ☒ Clear the aircraft via the filed route, if possible
 - ☐ Instruct the pilot to hold until able to comply
 - ☐ Lecture the pilot for slowing down traffic
- What are the two types of departure procedures? (Select the correct answer.)
 - ☐ STARs and SIDs
 - ☐ Vector and Pilot Navigational
 - ☒ ODPs and SIDs

Standard Instrument Departure (SID) Charts Summary

Standard Instrument Departures are primarily designed for system enhancement to expedite traffic flow and to reduce pilot/controller workload and frequency congestion.

EXERCISE – USING A SID

Purpose

This exercise will give you practice in using a Standard Instrument Departure (SID) chart.

Directions

Working in teams, read the scenario, then use the sample SID to locate the answers to the questions. After completion, select a spokesman for your team to provide and justify your answers.

Reference – Using a SID

Scenario

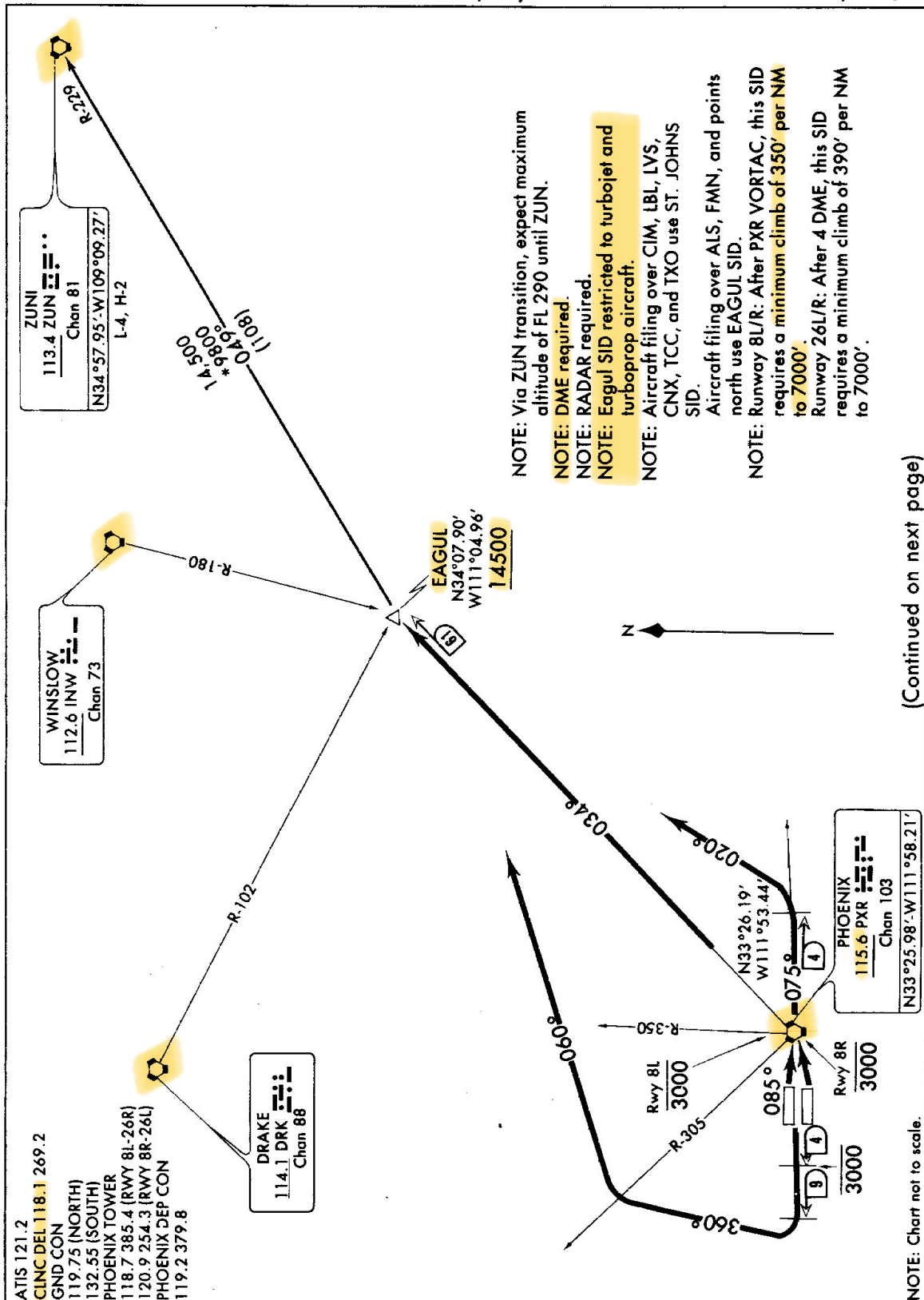
The pilot of N100WN, a GLF3, is preparing to depart Phoenix Sky Harbor Airport for Chicago O'Hare. ATIS has advised N100WN that Runway 26R is active. The clearance delivery controller has cleared N100WN "...via the EAGUL THREE departure, Zuni transition."

Question	Answer
1. What type of SID is the EAGUL THREE departure?	Pilot Nav
2. Can a Cessna 172 aircraft fly the EAGUL THREE departure? Explain.	No, restricted to turboprop/turbopump
3. How many VORTACs are displayed on the EAGUL THREE departure?	4
4. What navigational frequency will the pilot of N100WN have selected as the aircraft is taking off from Sky Harbor Airport?	115.6
5. Must an aircraft have DME to execute the EAGUL THREE departure? Explain.	Yes, notes require DME
6. What is the three-letter designator for Sky Harbor Airport?	PHX
7. On what frequency will a SID be issued?	118.1 Clearance Delivery
8. Where does the EAGUL THREE departure route terminate?	EAGUL
9. What is the altitude restriction at EAGUL Intersection? Is this AGL or MSL?	14500 MSL
10. What is the minimum climb rate for this scenario?	350 ft /NM
11. Who is primarily responsible for navigation on this SID?	Pilot

(EAGUL3.EAGUL) 99140

EAGUL THREE DEPARTURE

PHOENIX SKY HARBOR INTL (PHX)
PHOENIX, ARIZONA



(Continued on next page)

EAGUL THREE DEPARTURE
(EAGUL3.EAGUL) 99140

PHOENIX SKY HARBOR INTL (PHX)
PHOENIX, ARIZONA

STANDARD TERMINAL ARRIVAL (STAR) CHARTS

Purpose: The purpose of this lesson is to describe how Standard Terminal Arrival (STAR) charts are used and to identify the types of information displayed on these charts.

Objectives:

- Identify the purpose of STARs
- Identify layout/components of STARs
- Identify how to interpret a STAR
- Identify “No STAR” procedures

References for this lesson are as follows:

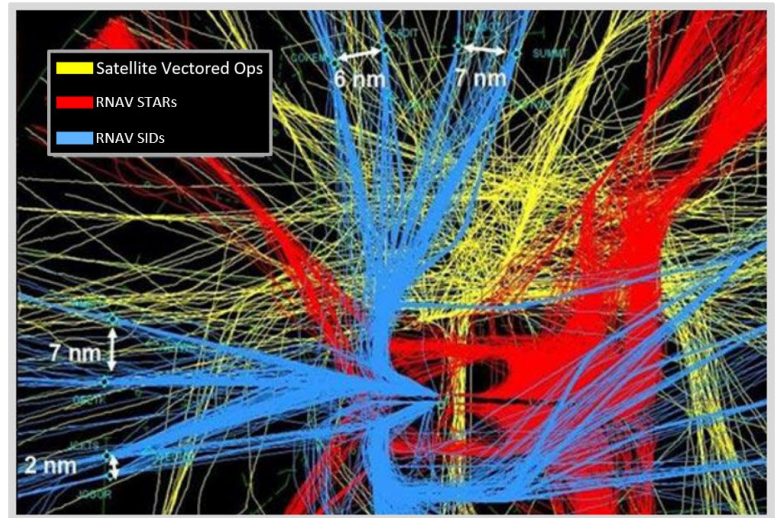
- FAA Order JO 7110.65, Air Traffic Control
- FAA-H-8083-16, Instrument Procedures Handbook
- U.S. Terminal Procedures Chart SW, Vol. 1

Performance-Based Navigation (PBN)

Like the Standard Instrument Departure (SID), the STAR uses Performance-Based Navigation (PBN). PBN provides transitions to a safer and more efficient design of National Airspace System (NAS).

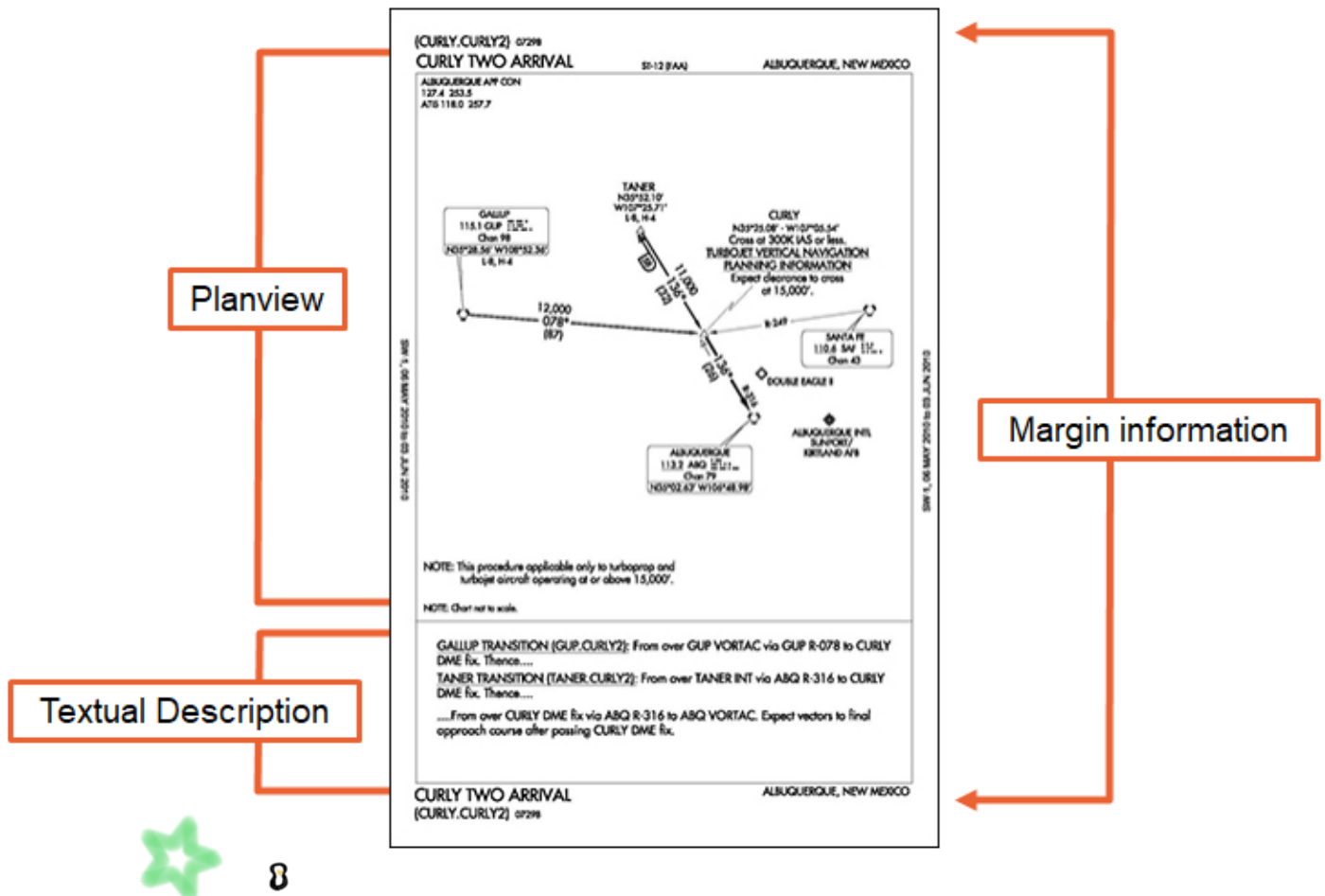
PBN impacts arrivals:

- Reduces arrival delay times at the destination airports
- Improves traffic flow management
- Improves access to more airports during poor weather conditions



Standard Terminal Arrival (STAR) Charts

Like the SID, the STAR has three sections on a chart.

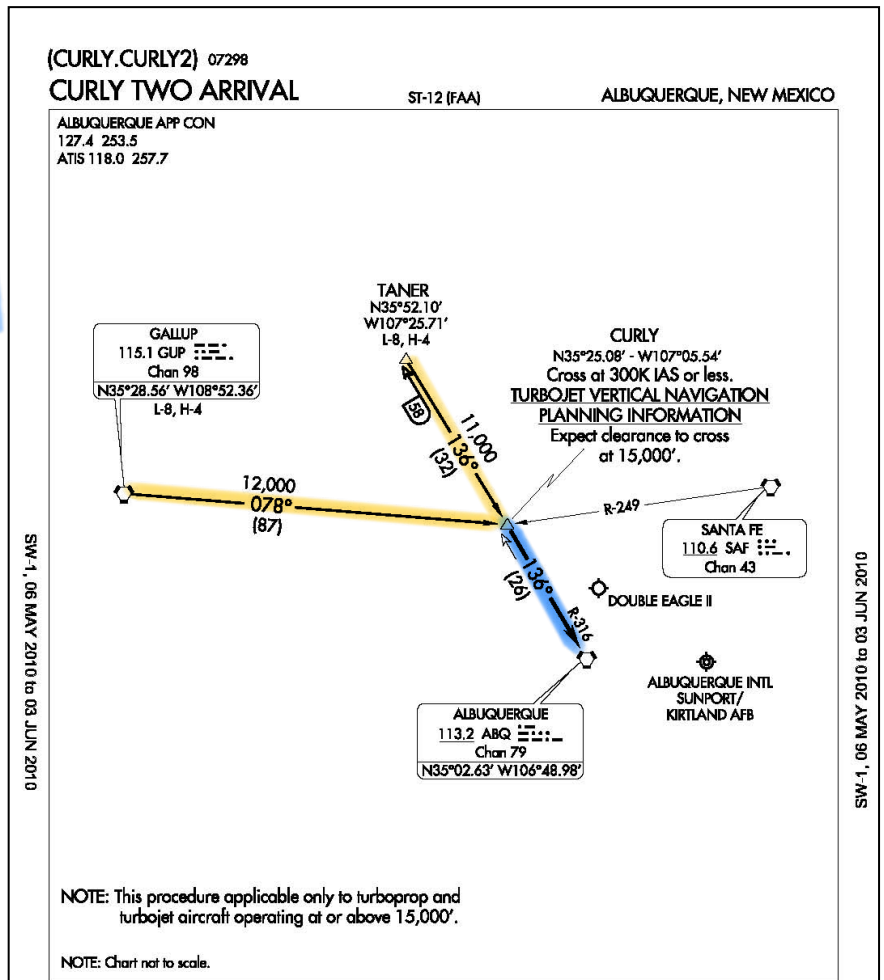


STARs are long-range planning tools that funnel arrival traffic into one-way corridors. They are designed to expedite arrival procedures and to facilitate transition between en route and instrument approach operations.

- They depict preplanned IFR arrival procedures in graphic and textual form
- Each STAR procedure is presented as a separate chart and may serve either a single airport or more than one airport in a given geographic area
- Use of a STAR requires pilot possession of the approved chart

Procedure:

- The transition routes begin at GUP VORTAC or TANER DME fix
- Transition routes (medium black lines) provide navigation to the common fix, CURLY (for which the procedure is named)
- The arrival route (thick black line) begins at CURLY and ends at ABQ VORTAC
- Frequencies: Navigation and Communication



Standard Information:

- The communication frequencies for the procedure are listed at the top left of the STAR; the navigation frequencies are listed in the individual navigational aid information box
- Turbojet Vertical Navigation Planning Information
 - These are "expect" clearances and not to be executed until the controller issues clearance.
- Textual Description
- Procedure Notes

GALLUP TRANSITION (GUP.CURLY2): From over GUP VORTAC via GUP R-078 to CURLY DME fix. Thence....

TANER TRANSITION (TANER.CURLY2): From over TANER INT via ABQ R-316 to CURLY DME fix. Thence....

....From over CURLY DME fix via ABQ R-316 to ABQ VORTAC. Expect vectors to final approach course after passing CURLY DME fix.

CURLY TWO ARRIVAL
(CURLY.CURLY2) 07298

ALBUQUERQUE, NEW MEXICO

U.S. Terminal Procedures Chart – STARS

Unlike SIDs, which are located with the Instrument Approach Procedures (IAPs) and the Airport Diagram, STARS have their own separate section in the front of the book.

INDEX OF TERMINAL CHARTS AND MINIMUMS							
NAME	PROC	SECT	PG	NAME	PROC	SECT	PG
AKRON, CO				ALBUQUERQUE, NM			
COLORADO PLAINS RGNL (AKO)				ALBUQUERQUE INTL SUNPORT (ABQ)			
TAKE-OFF MINIMUMS			C	TAKE-OFF MINIMUMS			C
ALTERNATE MINIMUMS			E	ALTERNATE MINIMUMS			E
IAPS..... RNAV (GPS) RWY 11			1	RADAR MINIMUMS			N
RNAV (GPS) RWY 29			2				
VOR RWY 29			3				
ALAMAGORDO, NM				STARS.... CURLY TWO			
ALAMAGORDO-WHT SANDS RGNL(ALM)				FRIHO FOUR			
TAKE-OFF MINIMUMS			C	LAVAN THREE			
IAPSRNAV (GPS) RWY 3			4	MIERA TWO			
VOR/DME RWY 3			5	IAPS ILS OR LOC RWY 3			13
VOR RWY 3			6	ILS OR LOC RWY 8			14
DPS.....CORONA ONE (OBSTACLE)			7	RNAV (GPS) RWY 3			15
				RNAV (GPS) RWY 8			16
				RNAV (GPS) RWY 17			17
				RNAV (GPS) RWY 35			18
ALAMAGORDO, NM							
---SEE HOLLOMAN AFB							

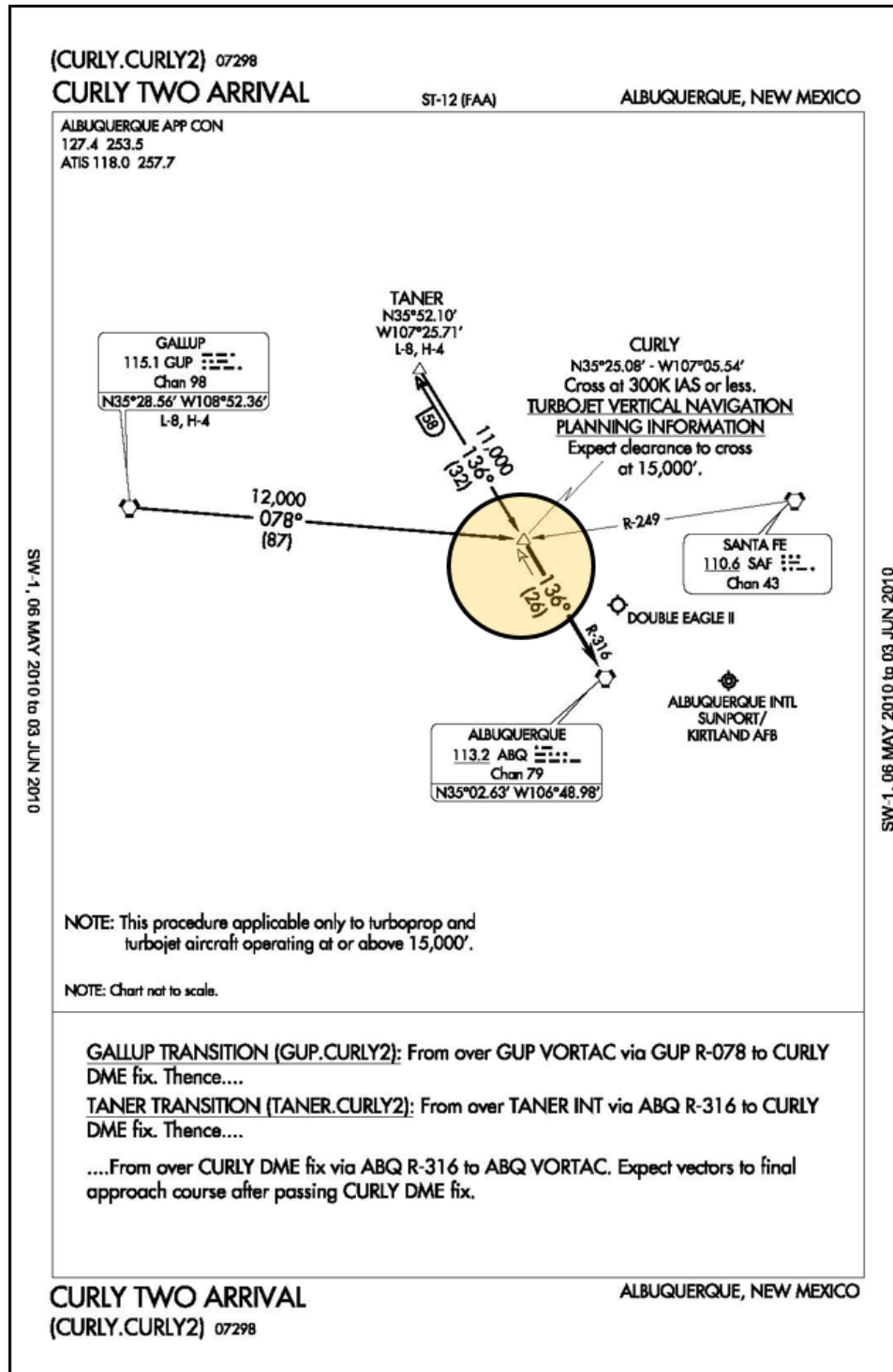
Interpreting a STAR

STARs use much of the same symbology as departure and approach charts.

Common NAVAID/Fix

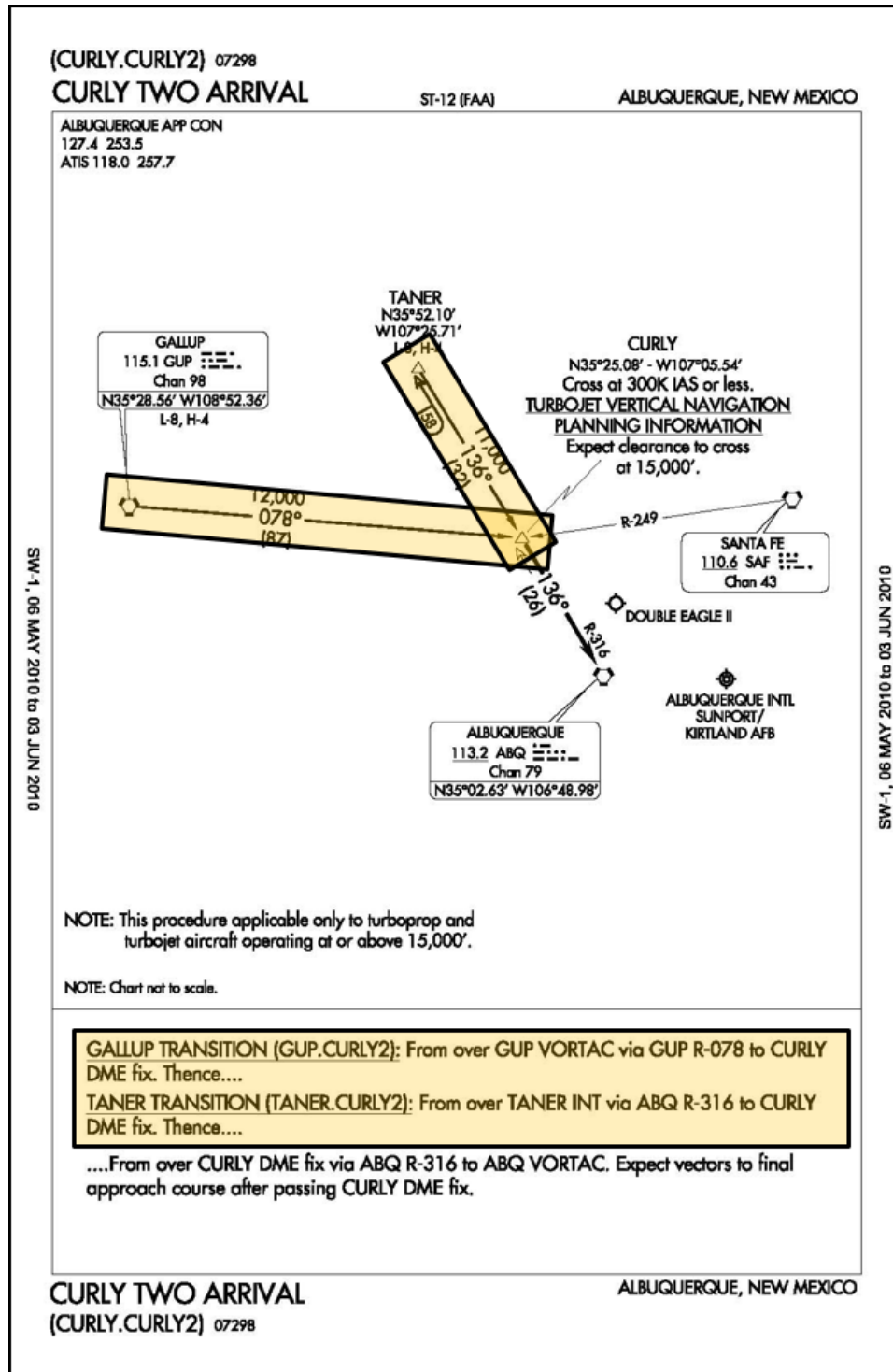
The arrival route officially begins at the common NAVAID, intersection, or fix where all the various transitions come together.

8



Transitions

- A transition is a published segment used to connect one or more en route airways, jet routes, or area navigation (RNAV) routes to the common NAVAID/fix
- Several transitions from various directions can be accommodated on the same chart, and traffic flow is routed appropriately within the congested airspace



Although it is **NOT** required, pilots are encouraged to include the phrase “No STAR” on their flight plan if they do not wish to be issued a STAR. The flight progress strip will reflect this decision.

Flight Progress Strip

Example

A clearance without using a STAR may read:	The same clearance using a STAR would read:
“Learjet 32G, cleared to Seattle/Tacoma International Airport as filed. Maintain 12,000. At the Ephrata VOR, intercept the 221° radial to CHINS Intersection. Intercept the 284° radial of the Yakima VOR to RADDY Intersection. Cross RADDY at 10,000. Continue via the Yakima 284° radial to AUBRN. Expect radar vectors to the final approach course.”	“Learjet 32G, cleared to Seattle/Tacoma International Airport as filed, then CHINS EIGHT ARRIVAL, Ephrata Transition. Maintain 10,000.” A shorter transmission conveys the same information.



Knowledge Check B

REVIEW what you have learned so far about STAR charts. ANSWER the questions listed below.

1. What are the purposes of a STAR chart? *(Select all correct answers that apply.)*
 - ☐ Expedite ATC departure procedures
 - ☐ Transition from approach to terminal
 - ☒ Expedite ATC arrival procedures
 - ☒ Transition aircraft from en route to approach
2. What types of information will you find on a STAR? *(Select all correct answers that apply.)*
 - ☒ Multiple converging transition routes
 - ☒ One common fix, intersection, or NAVAID
 - ☒ Transitions from en route to approach
 - ☒ A description of the Planview information
3. What are pilots encouraged to include on their flight plan if they do not wish to be issued a STAR? *(Select the correct answer.)*
 - ☐ "STAR exempt"
 - ☒ "No STAR"
 - ☐ "STAR not issued"

Standard Terminal Arrival (STAR) Summary

Standard terminal arrivals streamline arrivals by allowing aircraft to merge to a given point from one or more directions and land safely on a runway. Controllers must become familiar with standard terminal arrivals in order to provide successful guidance to arriving aircraft.

EXERCISE – USING A STAR

Purpose

This exercise will give you practice in using a Standard Terminal Arrival (STAR) chart.

Directions

Working in teams read the scenario, then use the sample STAR to locate the answers to the questions. After completion, select a spokesman for your team to provide and justify your answers.

Reference – Using a STAR

Scenario

The pilot of N1GN, a B737, is en route to Las Vegas, Nevada, from New York City. Somewhere over Nebraska at FL 280, the pilot receives a clearance “Direct PGS (Peach Springs) VEEVA TWO ARRIVAL to LAS (Las Vegas).”

Question	Answer
1. Normally, what type of controller cleared N1GN for the VEEVA TWO Arrival?	Enroute
2. When N1GN received the VEEVA TWO Arrival clearance, what type of navigational chart was the pilot using?	Enroute High Alt.
3. What must a pilot possess before accepting a STAR clearance?	The STAR chart
4. Why isn't the altitude information at VEEVA included in the textual description?	Included in planview
5. After receiving the clearance “Direct PGS the VEEVA TWO ARRIVAL to LAS,” what frequency would the pilot select to navigate direct PGS?	112.0
6. To find out the LAS weather, approach-in-use, and active runway, what frequency would the pilot select?	132.4
7. How can a pilot identify VEEVA intersection?	VORTAC, @ Boulder City & Needles

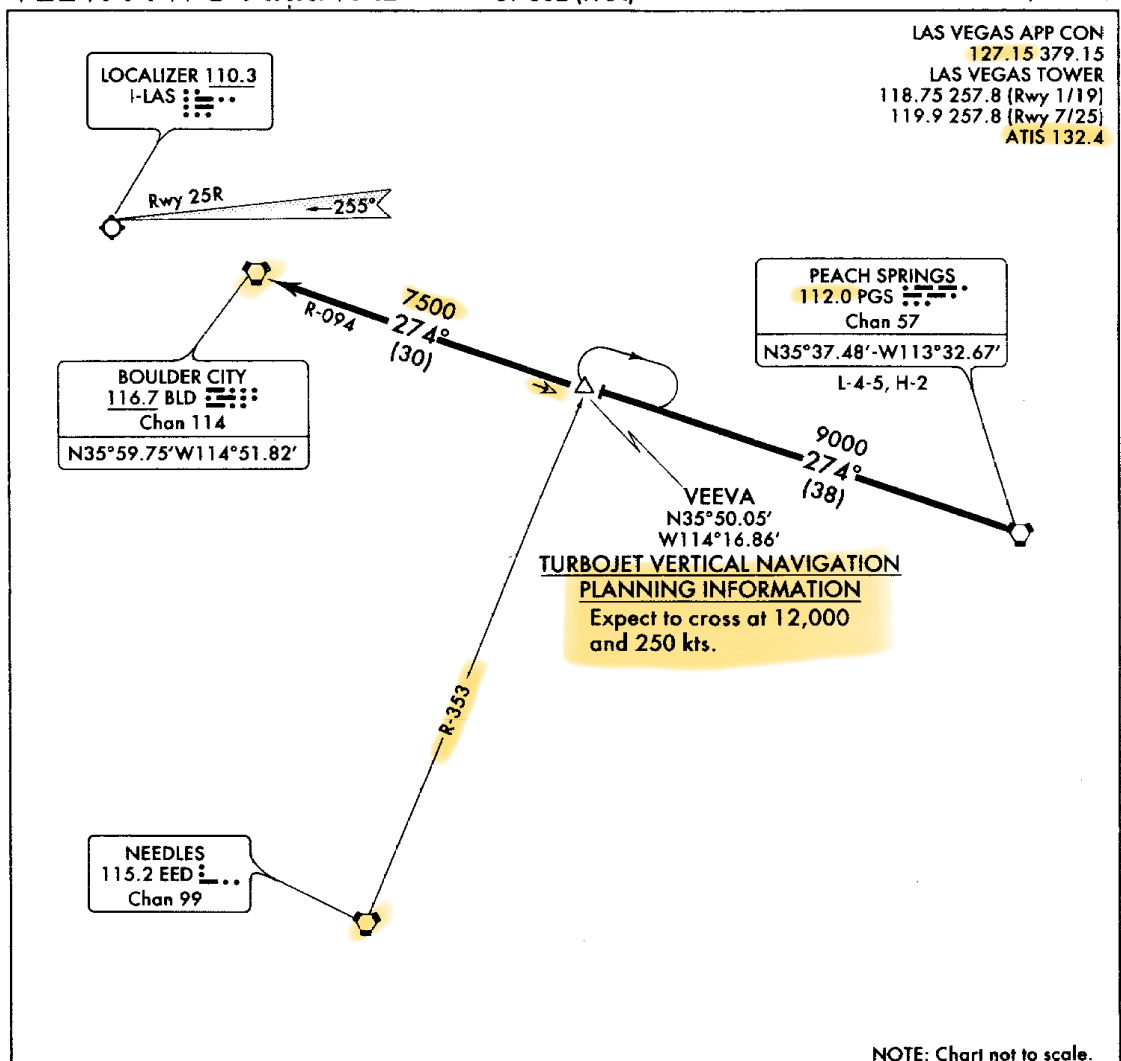
Question	Answer
8. Where does the VEEVA TWO ARRIVAL terminate?	Boulder City VORTAC
9. What is the lowest usable altitude that N1GN can be assigned on the VEEVA TWO ARRIVAL? Is this MSL or AGL?	7500 MSL
10. May the Needles transition be used if desired?	No, This line = Radial, NOT transition
11. What frequency will the last en route controller assign to N1GN?	127.15
12. If the active runway at LAS is RWY 19L, what will N1GN do after BLD VORTAC?	Expect Vectors to final approach

(PGS.VEEVA2) 98057

VEEVA TWO ARRIVAL

ST-662 (FAA)

McCARRAN INTL
LAS VEGAS, NEVADA



From over PGS VORTAC via PGS R-274 and BLD R-094 to BLD VORTAC. Runways 25L/R expect ILS Approach Procedure. Runways 1L/R, 7L/R, and 19L/R expect vectors to final approach course.

VEEVA TWO ARRIVAL
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SUMMARY

The purpose of this module was to introduce you to Standard Instrument Departure (SID) and Standard Terminal Arrival (STAR) procedural charts, which are used by pilots when flying on an Instrument Flight Rules (IFR) flight plan.

In accordance with FAA Order JO 7110.65, Air Traffic Control; FAA-H-8083-16 Instrument Procedures Handbook; U.S. Terminal Procedures Chart SW, Vol. 1; and the Aeronautical Information Manual (AIM); you should now be able to:

- Identify types of departure procedures
- Identify types of SIDs
- Identify components of SIDs
- Identify “No SID” procedures
- Identify the purpose of STARs
- Identify layout/components of STARs
- Identify how to interpret a STAR
- Identify “No STAR” procedures

KNOWLEDGE CHECK ANSWER KEY

Knowledge Check A

1. Instrument Departure Procedures; 2. City; 3. Clear the aircraft via the filed route, if possible; 4. ODPs and SIDs

Knowledge Check B

1. Expedite ATC arrival procedures, Transition aircraft from en route to approach; 2. Multiple converging transition routes; One common fix, intersection or NAVAID; Transitions from en route to approach; A description of the Planview information; 3. “No STAR”